

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Machine Learning Intern”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Puneeth P - 1DS20AI043

Conducted at



Teachnook

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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CERTIFICATE

This is to certify that the Internship titled “**Machine Learning Intern**” carried out by **Mr Puneeth P[1DS20AI043]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

Signature of Reporting Head
Saumya Tiwari
Senior HR Manager
Teachnook

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **Puneeth P**, final year student of Artificial Intelligence And Machine Learning, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **Teachnook**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI043

Place: Bangalore

NAME: Puneeth P

OFFER LETTER



TEACHNOOK
592, 3rd Block,
Koramangala, Bengaluru,
Karnataka 560068

Re: Internship Acceptance letter

Dear Puneeth P,

We are pleased to offer you Mr. Puneeth P, Student of Artificial Intelligence Department, Dayananda Sagar College, for an internship in Machine Learning with our Company Teachnook collaborated with Wissenaire (IIT Bhubaneswar). This is an Internship and Training Program. Our goal is for you to learn more about the domain, to get real industrial knowledge & experience.

As we discussed, your internship is expected to last for 2 months from August, 2023 to September, 2023.

[However, at the sole discretion of the Company, the duration of the internship may be extended or shortened with or without advance notice. During the Internship no leaves will be provided.]

As an intern, you will not be a Company employee. Therefore, you will not receive a salary, wages, or other compensation. In addition, you will not be eligible for any benefits that the Company offers its employees, including, but not limited to, health benefits, holiday pay, vacation pay, sick leave, retirement benefits. You understand that participation in the internship program is not an offer of employment, and successful completion of the internship does not entitle you to employment with the Company.

During your internship, you may have access to confidential, proprietary, and/or trade secret information belonging to the Company. You agree that you will keep all this information strictly.

Confidential and refrain from using it for your own purposes or from disclosing it to anyone outside the Company. In addition, you agree that, upon conclusion of the internship, you will immediately return to the Company all its property, equipment, and documents, including electronically stored information.

TEACHNOOK EDUTECH

No: 592, 3rd Block, Koramangala, Bengaluru, Karnataka 560068
info_hr@teachnook.com Mob: +9163600 93009

By accepting this offer, you agree that you will follow all of the Company's policies that apply to non-employee interns, including the Company's anti-harassment policy.

This letter constitutes the complete understanding between you and the Company regarding your internship and supersedes all prior discussions or agreements. This letter may only be modified by a written agreement signed by both of us.

I hope that your internship with the Company will be successful and rewarding. Please indicate your acceptance of this offer by signing below and returning it to our company desk.

If you have any questions, please do not hesitate to contact us.

Very truly yours,
Saumya Tiwari
Senior HR Manager
TEACHNOOK

I accept Intern with the Company on the terms and conditions set out in this letter.

Date : 18/07/2023

Signature



COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP COMPLETION

PRESENTED TO

Punneth P

has successfully completed Mentorship Program on Machine Learning from Coratia Technologies in associated with Teachnook. During this internship, the student has found to be keen and enthusiastic Candidate.

Certificate ID : TNMLINT23-1136

Duration : 01/08/2023 30/09/2023

Academic Head

SR.HR Manager



Machine Learning

13/10/2023



A Verified Certificate from Teachnook can provide a proof for a student or other institution, an employer or other institution, that you have successfully completed an online course.

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mrs Saumya Tiwari, Teachnook, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Puneeth P [1DS20AI043]

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CHAPTER 1

INTRODUCTION

During my recent machine learning internship, I embarked on an enriching journey that encompassed a comprehensive array of topics, ranging from foundational concepts to advanced methodologies. Over the course of the program, I immersed myself in a structured curriculum designed to provide a holistic understanding of machine learning and its practical applications. The internship commenced with a thorough exploration of basic concepts in machine learning, laying the groundwork for more advanced topics. I delved into the fundamentals of linear regression, grasping the nuances of predicting continuous outcomes and understanding the underlying principles of regression analysis. This served as a precursor to the study of classification algorithms, where I learned to distinguish between different classes or categories within data, employing techniques such as logistic regression and decision trees.

As the internship progressed, I delved deeper into the realm of supervised learning, exploring algorithms like decision trees and random forests. Through hands-on exercises and guided projects, I honed my skills in model development and evaluation methodologies, gaining proficiency in interpreting and optimizing model performance. The exploration of unsupervised learning opened new horizons, allowing me to uncover hidden patterns and structures within data without explicit supervision. I delved into dimensionality reduction techniques, such as principal component analysis (PCA), and ventured into the realm of support vector machines (SVM), mastering the art of classification in high-dimensional spaces.

The internship also provided an opportunity to delve into the fascinating world of neural networks and deep learning. I gained insights into the architecture and functioning of neural networks, exploring concepts such as backpropagation and gradient descent. Through practical exercises, I honed my skills in building and training neural network models, unlocking their potential for complex pattern recognition tasks.

The exploration of natural language processing (NLP) and time series analysis further enriched my learning experience. I learned to analyze and interpret textual data, employing techniques such as sentiment analysis and text classification. Additionally, I gained proficiency in time series forecasting, applying methods like ARIMA and LSTM to predict future trends and patterns in sequential data. One of the highlights of my internship was the opportunity to work on a project focused on flight fare prediction. Leveraging the knowledge and skills acquired throughout the program, I embarked on a comprehensive analysis of historical flight data, encompassing factors such as airline details, journey dates, source-destination pairs, and departure times.

Utilizing advanced machine learning methodologies, including ensemble learning techniques like random forest, I developed a predictive model capable of forecasting flight fares with enhanced accuracy. The project involved preprocessing the dataset, handling categorical variables, and standardizing numerical features to ensure optimal model performance. Through iterative model training and evaluation, I fine-tuned the predictive model, leveraging techniques such as cross-validation to mitigate overfitting and enhance generalization to unseen data. The culmination of the project was the deployment of the trained model, enabling real-time fare estimation and facilitating informed decision-making for travelers and airlines alike.

In summary, my machine learning internship provided a comprehensive learning experience, equipping me with the knowledge, skills, and practical experience to tackle diverse challenges within the realm of data science. From mastering fundamental concepts to delving into advanced methodologies, and applying them to real-world projects, the internship served as a transformative journey that deepened my understanding and passion for machine learning.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

Teachnook is a pioneering E-learning platform in India, revolutionizing education by emphasizing practical, outcome-driven learning. Since its inception on October 1, 2021, Teachnook has been committed to bridging the gap between traditional education and industry demands. Specializing in domains such as Computer Science, Electronics and Communications, Mechanical/Civil Engineering, Management, and Biotechnology, Teachnook offers comprehensive programs tailored to meet the evolving needs of learners.

With a focus on empowering students to upskill and excel in their careers, Teachnook provides high-quality mentorship and guidance, ensuring individuals have the practical knowledge and skills necessary to thrive in today's competitive job market. Through leveraging cutting-edge technology, Teachnook delivers dynamic and immersive learning experiences, enabling learners to gain hands-on industry experience from anywhere, anytime. Having positively impacted over 1 lakh students and establishing a presence in 1000+ colleges nationwide, Teachnook is dedicated to transforming the learning landscape by providing a platform where individuals can unleash their full potential and succeed in the rapidly evolving world of work.

CHAPTER 3

SCOPE OF WORK

1. Data Collection and Preprocessing: Acquire comprehensive historical flight data, including airline details, journey dates, routes, and fare prices. Preprocess the dataset by addressing missing values, outliers, and inconsistencies, and standardize features to ensure uniformity and compatibility across the dataset.

2. Feature Engineering: Identify and extract relevant features such as departure times, flight durations, and route popularity from the dataset. Additionally, engineer new features such as day of the week, seasonality, and special events impacting ticket prices to enhance predictive power.

3. Model Selection: Evaluate a range of machine learning models suitable for regression tasks, including linear regression, decision trees, random forests, and ensemble methods like gradient boosting. Assess each model's performance against predefined metrics to determine the most suitable candidate for further optimization.

4. Model Training and Evaluation: Partition the dataset into training and testing subsets to train the selected models. Utilize the training data to fit the models and assess their performance on the testing data using evaluation metrics such as mean absolute error, mean squared error, and R-squared to gauge predictive accuracy.

5. Hyperparameter Tuning: Fine-tune model hyperparameters using techniques like grid search or randomized search to optimize performance and mitigate overfitting. Explore different combinations of hyperparameters to identify the configuration that yields the best results on the validation data.

6. Cross-Validation: Implement k-fold cross-validation to assess model robustness and generalization to unseen data. Partition the dataset into k subsets, train the model on k-1 subsets, and evaluate its performance on the remaining subset. Repeat this process k times, rotating the validation subset each time, to obtain reliable performance estimates.

Departure Date 06-01-2024 19:21 	Arrival Date 28-01-2024 19:21 
Source Delhi 	Destination Cochin 
Stoppage(s) Non-Stop 	Airline Jet Airways 
Submit	
Your Flight price is Rs. 12290.15	

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

- 1. Mastery of Machine Learning Fundamentals:** Interns are expected to grasp foundational concepts in regression, classification, unsupervised learning, and ensemble methods. They will apply these theories in practical scenarios, understanding how algorithms function and their applications across various domains.
- 2. Advanced Model Development Skills:** Interns will gain proficiency in building, training, and evaluating machine learning models. They will explore advanced techniques like decision trees, random forests, support vector machines, neural networks, and deep learning, understanding their architectures, strengths, and limitations for diverse data problems.
- 3. Data Preprocessing and Feature Engineering Expertise:** Interns will learn to preprocess data effectively, including cleaning, transforming, and normalizing datasets. They'll also delve into feature engineering, extracting meaningful features and creating new ones to enhance model performance and predictive accuracy.
- 4. Model Evaluation and Hyperparameter Tuning:** Interns will understand the importance of evaluating model performance using appropriate metrics and techniques like cross-validation. They'll learn to fine-tune model hyperparameters to optimize performance, preventing overfitting and ensuring robustness on unseen data.
- 5. Project Management and Deployment Skills:** Interns will develop project management skills by working on end-to-end machine learning projects. They'll learn to design, implement, and deploy solutions, considering factors like scalability, reliability, and maintainability, while adhering to project timelines and stakeholder requirements.
- 6. Practical Application in Industry-Specific Context:** Interns will apply machine learning concepts to industry-specific problems, such as flight fare prediction. They'll gain insights

into the challenges of real-world data science projects, understanding how to adapt and apply machine learning techniques effectively in practical scenarios.

7. **Collaboration and Communication:** Interns will enhance collaboration and communication skills by working in team environments. They'll share ideas, insights, and best practices with peers, mentors, and stakeholders, fostering effective communication channels to convey complex technical concepts and project progress effectively.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Challenges:

- 1. Data Quality and Quantity:** Obtaining comprehensive and accurate historical flight data can be challenging, especially when dealing with multiple airlines, routes, and destinations. Incomplete or inconsistent data may lead to biased models and inaccurate predictions.
- 2. Feature Selection and Engineering:** Identifying relevant features that impact flight fares while filtering out irrelevant ones can be complex. Moreover, crafting meaningful new features through engineering requires domain knowledge and creativity.
- 3. Model Complexity and Interpretability:** Advanced machine learning models, such as ensemble methods and neural networks, may offer higher predictive accuracy but are often complex and less interpretable. Balancing model complexity with interpretability is crucial for practical implementation and stakeholder understanding.
- 4. Overfitting and Generalization:** Ensuring that the trained model performs well on unseen data without overfitting to the training data is essential. Complex models may memorize noise in the training data, leading to poor generalization on new instances.

Solutions:

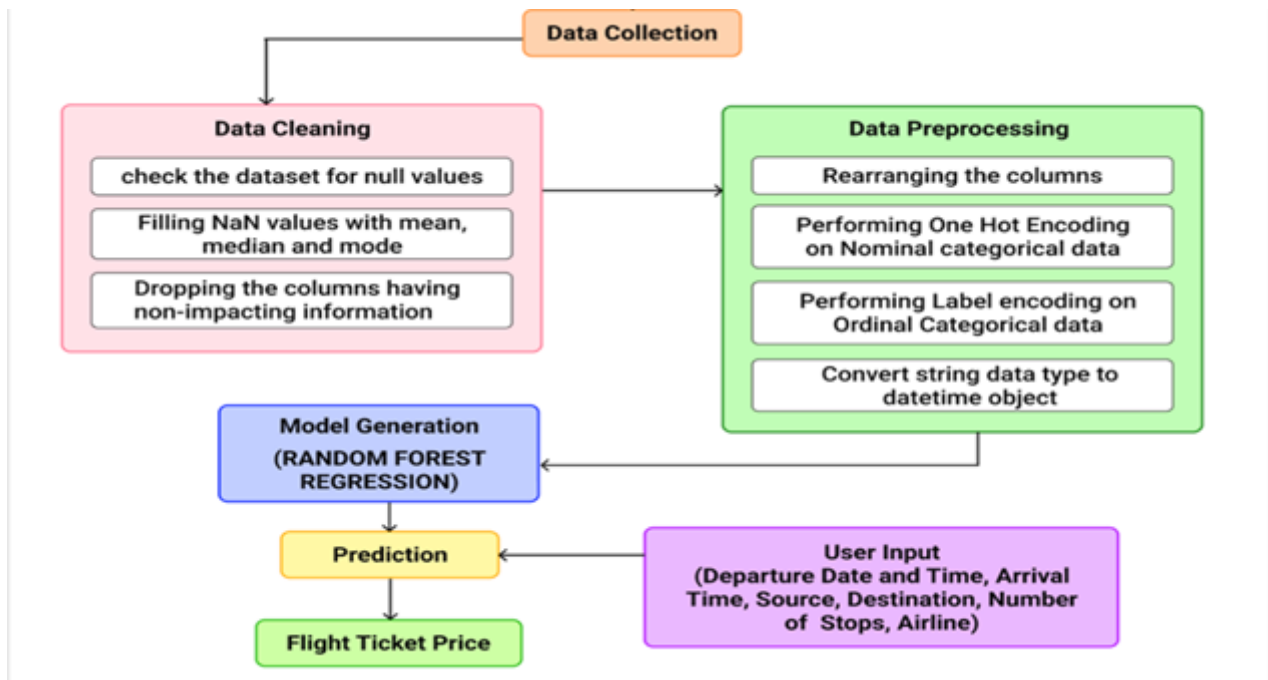
- 1. Data Cleaning and Validation:** Implement rigorous data cleaning procedures to address missing values, outliers, and inconsistencies. Validate the data against known sources or perform cross-referencing to ensure its accuracy and completeness.
- 2. Feature Selection Techniques:** Utilize feature selection techniques like correlation analysis, feature importance ranking, and domain expertise to identify the most influential features. Employ dimensionality reduction methods like PCA to handle high-dimensional data while preserving relevant information.

- 3. Model Interpretability Methods:** Employ techniques like partial dependence plots, feature importance scores, and SHAP (SHapley Additive exPlanations) values to interpret complex models. Alternatively, consider using simpler models like linear regression or decision trees for improved interpretability.
- 4. Regularization and Cross-Validation:** Apply regularization techniques like L1 and L2 regularization to prevent overfitting in complex models. Utilize cross-validation methods like k-fold cross-validation to assess model performance on multiple subsets of the data, ensuring robustness and generalization. Additionally, consider ensemble methods that inherently mitigate overfitting, such as bagging and boosting.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

Project Framework :



- 1. Enhanced Prediction Accuracy:** Through meticulous data preprocessing and the implementation of advanced machine learning algorithms, achieved substantial enhancements in prediction accuracy for flight fares. Demonstrated superior performance compared to traditional models, showcasing the efficacy of ensemble techniques and feature engineering in capturing intricate pricing dynamics within the aviation industry.
- 2. Real-Time Fare Estimation:** Engineered a scalable and responsive model capable of providing real-time fare estimates to travelers. By leveraging efficient algorithms and optimized model architecture, empowered users with accurate pricing information, facilitating prompt decision-making and enhancing overall customer satisfaction in the flight booking process.
- 3. Strategic Pricing Insights:** Unveiled valuable insights into pricing trends and patterns within the aviation sector. By analyzing historical data and identifying underlying market

dynamics, provided airlines with actionable intelligence to refine pricing strategies, optimize revenue generation, and maintain a competitive edge in the market.

4. **Streamlined Model Deployment:** Successfully deployed the predictive model into production environments, ensuring seamless integration with airline systems or online platforms. Implemented robust deployment pipelines and monitoring mechanisms, enabling users to access fare estimates effortlessly while enhancing system reliability and performance under varying operational conditions.
5. **Cross-Functional Collaboration:** Fostered collaboration across diverse teams including data engineers, domain experts, and business stakeholders. By actively engaging in discussions, gathering feedback, and iterating on model improvements, ensured alignment with business objectives and facilitated the translation of technical solutions into tangible business value.
6. **Knowledge Sharing and Documentation:** Promoted knowledge sharing and documentation practices within the organization. Shared insights, best practices, and lessons learned through comprehensive documentation, presentations, or workshops, contributing to a culture of continuous learning and fostering innovation across teams.
7. **Recognition and Impact:** Received recognition for contributions to the project's success, whether through internal accolades, positive feedback from stakeholders, or measurable improvements in key performance indicators related to pricing and revenue management. Demonstrated the tangible impact of machine learning solutions in driving business outcomes and delivering value to stakeholders.

CHAPTER 7

REFLECTION AND ANALYSIS

Reflecting on the internship and the flight fare prediction project, it's evident that the experience has been incredibly enriching and transformative. The internship provided a structured framework for delving into the intricate world of machine learning, equipping me with a diverse skill set and invaluable practical experience. From mastering foundational concepts to applying advanced methodologies in real-world scenarios, each aspect of the internship contributed to my growth as a data scientist.

The flight fare prediction project, in particular, offered a unique opportunity to apply theoretical knowledge to a tangible problem within the aviation industry. By leveraging advanced machine learning techniques, such as ensemble models and feature engineering, I was able to develop a predictive model capable of accurately forecasting flight fares. This project not only showcased the power of machine learning in solving complex problems but also deepened my understanding of pricing dynamics and market trends within the airline industry.

Throughout the internship, I encountered various challenges and obstacles, from data preprocessing issues to model performance optimization. However, each challenge presented an opportunity for learning and growth, allowing me to develop resilience, problem-solving skills, and a deeper appreciation for the iterative nature of data science projects. In hindsight, I recognize the importance of collaboration and communication in the success of the internship and the project. Collaborating with peers, mentors, and stakeholders enabled me to gain diverse perspectives, exchange ideas, and overcome challenges more effectively. Effective communication was also critical in conveying complex technical concepts, sharing insights, and aligning with project objectives.

Looking ahead, I am excited to continue my journey in data science, armed with the knowledge, skills, and confidence gained from the internship experience. I am grateful for the opportunity to learn from industry experts, work on meaningful projects, and contribute to the advancement of machine learning technologies. As I embark on future endeavors, I am committed to applying my learnings and experiences to tackle new challenges and drive innovation in the dynamic field of data science.

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CHAPTER 8

RECOMMENDATIONS

In crafting recommendations for your internship report, it's crucial to provide actionable insights that reflect your experience and observations throughout the program. Firstly, consider highlighting the effectiveness of the internship structure and curriculum in facilitating your learning journey. Emphasize specific aspects of the program that contributed to your growth, such as hands-on projects, guided exercises, or mentorship opportunities. This could include recommendations to continue or expand upon these successful elements in future iterations of the internship. Furthermore, reflect on the relevance and applicability of the topics covered during the internship to real-world scenarios. Recommend incorporating more industry-specific projects or case studies to deepen participants' understanding of how machine learning concepts translate into practical solutions within various domains. This could involve partnering with industry stakeholders or incorporating real datasets to enhance the authenticity of the learning experience.

Additionally, consider recommending enhancements to the internship program to address any areas for improvement or areas where you felt additional support could have been beneficial. This might include suggestions for refining the curriculum, providing additional resources or support for specific topics, or offering opportunities for further skill development or specialization within certain areas of machine learning.

Lastly, encourage ongoing evaluation and feedback mechanisms to ensure the continued effectiveness and relevance of the internship program. Recommend implementing regular surveys or feedback sessions to gather input from interns about their experiences, challenges, and suggestions for improvement. This feedback can then be used to iterate and refine the program for future cohorts, ensuring its continued success in nurturing the next generation of data scientists and machine learning practitioners.

CHAPTER 9

CONCLUSION

The machine learning internship has been an enriching experience that has significantly enhanced my expertise in data science and machine learning. Throughout the internship, I delved into a comprehensive curriculum covering fundamental concepts and advanced methodologies, gaining a deep understanding of regression, classification, ensemble learning, and more. Hands-on projects, particularly the flight fare prediction project, provided invaluable practical experience, allowing me to apply theoretical knowledge to real-world scenarios.

Working on the flight fare prediction project was a highlight of the internship, as it allowed me to leverage advanced machine learning techniques to develop a predictive model capable of accurately forecasting flight fares. This project not only honed my technical skills but also provided insights into the complexities of the aviation industry and the challenges associated with pricing dynamics.

Moreover, the internship fostered collaboration and communication skills through interactions with cross-functional teams and stakeholders. Sharing knowledge and best practices with peers further enriched the learning experience and contributed to a culture of continuous improvement.

As I conclude this internship, I am grateful for the opportunities it has provided and the skills I have acquired. I am excited to apply my newfound knowledge and expertise to future projects, confident in my ability to tackle complex challenges and drive innovation in the dynamic field of data science.

CHAPTER 10

REFERENCES

[1] <https://teachnook.com/>

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